They are,

1. BCD codes
2. Gray codes
3. Excess 3 code
4. ASCII code

8.6.1 BCD - 8421 Code Conversion

A group of bits (usually four) which are used to represent decimal numbers 0 to 9 are called BCD(Binary Coded Decimal) codes. The most popular BCD code is 8421 code. The 8421 indicates the binary weights of the four bits ($2^3, 2^2, 2^1, 2^0$). Using the four bits with weights 8,4,2,1, we can easily represent the decimal numbers 0 to 9 as given in the Table 8.3.

Table 8.3: Conversion of Decimal Number into BCD Code

Decimal Numbers	BCD Code
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	0001 0000
56	0101 0110
963	1001 0110 0011

8.6.2 Gray Code

The gray code is not a weighted code. Therefore it is not suitable for arithmetic operations, but finds applications in input/output devices and in some types of analog to digital converters.

Table 8.4: Gray code conversion

Decimal numbers	Binary code	Gray code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

The gray code is a minimum change code in which only one bit in the code group changes when moving from one step to the next. The gray code is also called as reflected binary code, which has a special property of containing two adjacent code numbers that differ by only one bit. The gray code representation for the decimal numbers 0 to 15, together with the binary code is given in the Table 8.4.

8.6.3 Excess-3 Code

The excess-3 code is another BCD code used in earlier computers. The excess-3 code is not a weighted code. It is a self-complementing code and helps in performing subtraction operations in digital computers. The excess–3 code is also a reflection code.

An excess–3 code is obtained by adding 3 to each digit of a decimal number. For example, to encode the decimal number 6 into an excess–3 code, we must first add 3, in order to obtain 9. The 9 is then encoded into its equivalent 4 bit binary code 1001.

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